

Homework — 1.3.4 Web Technologies — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Part A (14 marks)

1. (a) [1] The element/tag selector (selecting by the tag name `p`) — it applies to all `<p>` elements.
- (b) [2] 1 mark: an `#id` selector. 1 mark: because there is only **one** warning box, and an id is unique to a single element on the page (a class would be used if the same style applied to several elements). (Accept a class with justification only if the student argues it may be reused, but id is the better fit for a single element.)
2. [2] Any two for 1 mark each: the styling is defined once and applied across many/all pages, so the look is consistent; a single change to the CSS updates every page (easier maintenance); it separates content (HTML) from presentation (CSS); smaller/cleaner HTML and faster updates. (Max 2.)
3. [2] 1 mark: a web crawler/spider automatically visits/reads web pages and follows the links it finds to discover further pages. 1 mark: the content/keywords found are passed to the search engine which builds an index mapping keywords to the pages they appear on (for fast retrieval/ranking).
4. (a) [2] 1 mark: each incoming link counts as a "vote" for the page's importance. 1 mark: a vote/link from a high-ranking page carries more weight / passes on more rank than one from a low-ranking page, so it raises the target page's rank more.
- (b) [2] 1 mark: iterative means each page's rank is recalculated repeatedly using the latest rank values of the pages that link to it. 1 mark: converges means the process stops when the rank values stop changing significantly between iterations (settle to stable values).
5. (a) [1] Any one: instant feedback for the user without waiting for a server round trip / reduces load on the server because no request is needed for a simple check.
- (b) [2] 1 mark: only the server can compare the chosen username against the database of existing usernames (the browser does not hold that data). 1 mark: client-side code can be viewed/altered/bypassed, so the check is not trustworthy/secure on its own — it must run server-side where the user cannot tamper with it.

Part B (6 marks)

6. [2] 1 mark: a switch connects devices *within* a single LAN and forwards frames using **MAC** addresses. 1 mark: a router connects *different* networks (e.g. a LAN to the internet) and forwards packets using **IP** addresses.
7. [2] 1 mark: symmetric encryption uses the same secret key to encrypt and decrypt, so that key must somehow be shared with the other party in advance — if it is intercepted during sharing, security is lost. 1 mark: asymmetric uses a public/private key pair; the public key can be shared openly (only the matching private key, never shared, can decrypt), so no secret key needs to be exchanged.

8. [2] 1 mark for valid working (e.g. $45 = 32 + 8 + 4 + 1$, or repeated division by 2). 1 mark for the correct answer: **00101101**. (Award full marks for a correct answer with clear working; 1 mark only if working is correct but the final value has a minor error.)

Total: $14 + 6 = 20$ marks